MobileAgents: Designing and Evaluating a Mobile Multimodal Interface for Controlling Personal AI Agents On the Go

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CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Empirical studies in HCI**.

Keywords

multimodal interaction, agent orchestration, transparency, human-in-the-loop, trust, human–AI interaction

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1 Introduction: Use Case and Goals

AI Agents are increasingly capable [4]. However, these agents need to be controlled programmatically or through desktop applications. However, the value of these agents is that they can perform tasks while we are away, introducing demand for mobile agentic managers. This was demonstrated by ClawdBot [6], an AI agent controlled via messaging platforms. This underscores the need for a mobile agent platform that enables management of agents with adjustable levels of control and governance.

We designed a mobile-first platform that lets users *bring their own AI agents* and control them on the go through text, voice, and images (Figure 1). Text is sent via a standard conversational interface that supports asynchronous communication with agents. Voice enables hands-free control. Images allow you to send pictures from the phone camera.

Given that these workflows may execute tasks with varying impact and risks, we prioritize having a highly customizable level of control. Certain tasks will be executed completely autonomously, while others may require approval, additional input, or real-time monitoring.

This open-source application is initially targeted at technical people who use agents in their day-to-day lives. Examples may include academics, managing agents for explorative research, summarization, and documentation, or software engineers remotely steering teams of long-running coding agents.

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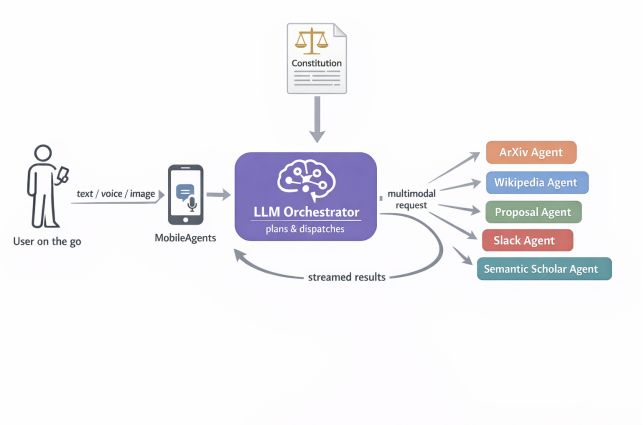


Figure 1: MobileAgents system overview. A user on the go issues a multimodal request (text, voice, or image) from their phone. The LLM Orchestrator decomposes it into a dependency-aware plan and dispatches tasks across the user's personal agent pool (ArXiv, Wikipedia, Proposal, Slack, Semantic Scholar). Results stream back to the mobile client in real time.

2 Methods: System Design and Implementation

2.1 Prototype Overview

MobileAgents is a fully functional prototype (Figure 1; full architecture in Appendix **??**–A).

2.2 Design Rationale

To begin, I drew on human-centred AI principles from the lectures (reliability, safety, transparency) [5] and looked online for interaction guidelines that emphasise: (a) setting expectations, (b) enabling efficient correction/repair, and (c) communicating uncertainty [1]. I also used Value Sensitive Design (VSD) [3] as inspiration into how to translate stakeholder values into concrete interaction requirements.

V1: Fast "It's a simple task; it should just do it instead of asking for permission."

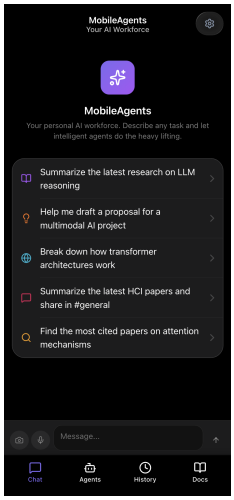
V2: Control "There should be guardrails and a control mechanism for sensitive tasks."

V3: Mobility: "I should be able to reach my agents hands-free."

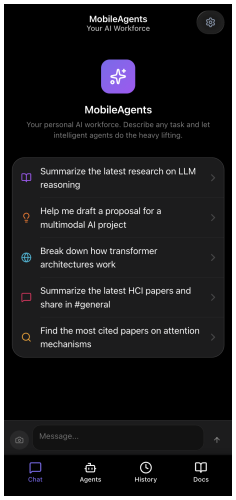
V4:

From these values, I drafted some initial resulting design implications, using methods from VSD [3] and related work [7]:

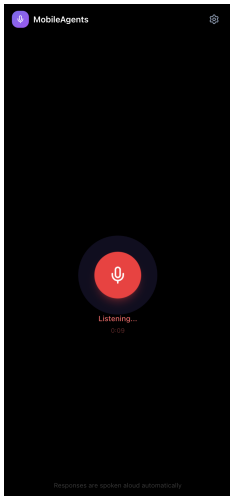
D1: Design a setting giving the agent full autonomy for low-r**D2:** Propose a plan and ask for confirmation before executing in plan mode; render a real-time execution graph showing active processes (Figure 3).



(a) Multimodal mode

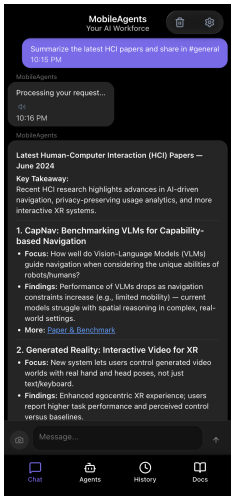


(b) Text + Images mode

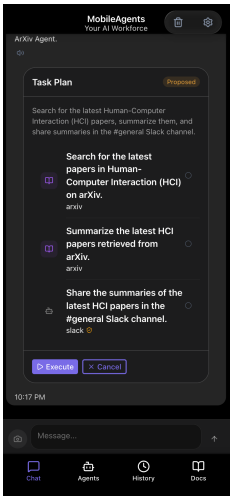


(c) Voice-only mode (listening)

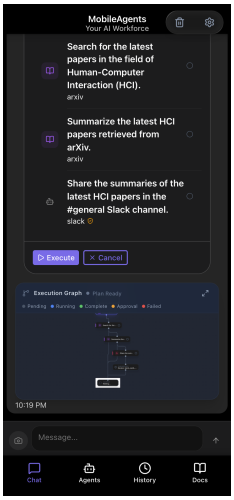
Figure 2: MobileAgents input modes. (a) Multimodal (b) Text + Images. (c) Voice-only



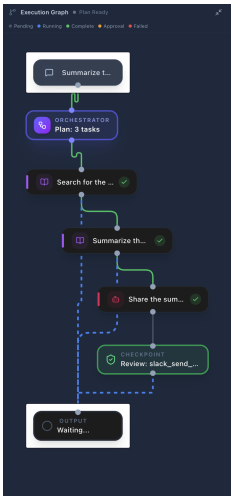
(a) C1: Black Box



(b) C2: Plan Preview



(c) C3: Full Transparency



(d) Live DAG (C3 detail)

Figure 3: The three transparency configurations and the C3 graph. (a) Black Box (b) Plan Preview (c) Full Transparency (d) Graph Close-up of Full Transparency

D3: Support customization of orchestration policy and output stylization (Appendix C).

D4: Provide the ability to turn agents on and off in the interface (Figure 4).

D5: Support hand-free voice-only mode (Figure 2c).

D6: Support images for quick snaps of environmental context (whiteboards, notes, etc) (Figure 2b).

I used these to begin thinking about the prototype, which I called 'Mobile Agents'. We support D1 and D2 via a control setting that includes Blackbox, Plan, and Transparency modes (Figure 3; Table 1). In. Blackbox gives the system full autonomy to execute tasks. Planning mode constructs a structured plan and asks the user for permission before executing. Transparency mode does everything Plan mode does and provides a real-time graph of execution traces for full visibility into the task. Additional control is implemented through D3 and D4. D3 is implemented through a natural-language constitution provito the Orchestrator agent in the system prompt to guide orchestration behavior and output style. D5 and D6 are exercised through the modality setting. We support text and images, native voice input/output (push-to-talk), and multimodal (speech-to-text, text, and images) (Figure 2). The text input is the baseline interaction with the multi-agent system and supports all control settings, mobile, and visualizations. Images are added to the text and are useful for capturing physical context (e.g., whiteboard, notes) using the device's camera. The voice-only setting enables hands-free use but does not support plan and transparency controls.

Multimodal is similar to text and images mode but also supports speech input that is converted to text, which does not limit plan and transparency modes.

Table 1: The three transparency configurations compared.

Configuration	What the user sees and can do
C1: Black Box	Agents auto-execute.
C2: Plan Preview	Approve orchestrator's task plan.
C3: Full Transparency	Plan with a real-time graph

3 Methods: Research Question, Evaluation Methods, and Metrics

3.1 Research Question

RQ1: Is there an efficiency/control tradeoff between the Blackbox, Plan, and Transparency modes?

RQ2: Does the support of multiple modalities (text, voice, images) improve the mobile-first experience?

RQ3: Are users able to tune the orchestration behavior through the constitution and on/off switches in the agent management tab (Figure 4)?

3.2 Study Design

The study runs in three sequential phases, each targeting one RQ. All phases use a within-subjects design; order effects are controlled by counterbalancing with a Latin square.

Phase 1 — Transparency modes (RQ1). Each participant completes the same research task (find and compare two recent papers on an assigned topic) under all three transparency configurations (Table 1)—Blackbox (C1), Plan Preview (C2), and Full Transparency (C3)—in counterbalanced order. The task is held constant so that any difference in time, approval behaviour, or subjective ratings is attributable to the configuration, not the task.

Phase 2 — Modality comparison (RQ2). Each participant completes a fixed multi-step workflow (find related work and draft a short proposal) using each of the three input modalities in counterbalanced order: (M1) text only, (M2) image (photograph a printed abstract), (M3) voice dictation (Figure 2). All three runs use Plan Preview (C2) as the transparency baseline so that modality is the only variable. This controls for configuration effects identified in Phase 1.

Phase 3 — Agentic customisation (RQ3). Participants receive an open-ended scenario: "Configure the system so that it never sends Slack messages without showing you a draft first, and remove any agents you would not want running automatically." They are then asked to run a multi-step query and observe whether the system respects their settings. This phase is not counterbalanced; it is a single exploratory session designed to surface whether participants can locate, understand, and apply the constitution editor and agent on/off toggles (Figure 4).

Tasks.

- **T1** (Phase 1 & 2 baseline): Type a text query to find and compare two recent papers on a given topic. Exercises ArXiv and Semantic Scholar agents.
- **T2** (Phase 2, M2): Photograph a printed paper abstract with the device camera and ask the system to find related work and draft a proposal. Exercises image analysis, ArXiv, Wikipedia, and Proposal agents.
- **T3** (Phase 2, M3): Use voice input to dictate a request to summarise results and send them to a Slack channel. Exercises voice transcription and the Slack agent with checkpoint approval.
- **T4** (Phase 3): Edit the orchestrator constitution and toggle at least one agent off, then rerun T1 and verify the change in behaviour.

3.3 Participants

We recruited 13 Cambridge students. All had experience with LLM-based tools. Participation was voluntary.

3.4 Evaluation Methods

In phase 1, participants completed a 7-point Likert survey for *perceived control* and *satisfaction* after each control setting. In phase 2, they also completed the NASA Task Load Index [**?**] to test for cognitive load across modalities, and the Single Ease Question [**?**] for usability. Participants rated the constitution and agent toggles on a 3-item perceived controllability scale. They shared live comments on the user experience as they tried out the application. An interview (questions in Appendix J)) explores transparency and modality preferences across RQ1 and RQ2. Additionally, it queries the usability and usefulness of agent controls for RQ3.

3.5 Metrics

RQ1 — Efficiency/control tradeoff:

- Trust, perceived control, and satisfaction (Likert, 1–7).
- Qualitative opinions on over- or under-informed cases.

RQ2 — Multimodal mobile experience:

- NASA-TLX composite score and subscale scores per modality.
- Single Ease Question (SEQ).
- Modality preference between different contexts.

RQ3 — Orchestration tuning:

- Task success after customisation: binary measure of whether the system respected the user's constitution rule and agent configuration during T4.
- Perceived controllability scale score (3-item Likert).
- Qualitatively on discoverability, understanding, and ownership.

4 Results

4.1 Quantitative Findings

Table 2 reveals that Trust and Control monotonically increasing from C1 to C3, which aligns with the increased level of human interactivity. In terms of satisfaction, C3 performed best (mean 5.8), and C2 performed worst (3.9), indicating users preferred either full autonomy or a high level of control. We compare the options sing

Table 2: Mean (SD) primary survey scores (1–7 Likert) per transparency configuration.

Metric	C1: Black Box	C2: Plan Preview	C3: Full Transp.
Trust	4.3 (1.7)	5.0 (1.5)	5.8 (1.0)
Control	3.2 (2.1)	5.4 (1.3)	5.7 (1.1)
Satisfaction	4.4 (2.3)	3.9 (1.1)	5.8 (1.1)

a pairwise Wilcoxon signed-rank test with a Bonferroni correction (Table 3). We do not find any significant differences in trust between any pairs. However, C2 and C3 are significant relative to C1 in terms of control. For satisfaction, only C3 has significant results over C2.

These results motivate the decision to make transparency mode the default setting for the app. However, all options remain available in the settings, as they appear to offer distinct benefits.

Table 3: Post-hoc pairwise Wilcoxon signed-rank tests (Bonferroni-corrected, $k = 3$) and Cohen's d for primary RQ1 metrics. $^{*}p_{\text{bon}} < .05$.

Metric	Pair	W	p_{bon}	d
Trust	C1-C2	16.0	0.756	-0.33
Trust	C1-C3	9.5	0.120	-0.69
Trust	C2-C3	18.0	0.621	-0.44
Control	C1-C2	6.0	0.041*	-0.88
Control	C1-C3	3.0	0.012*	-1.03
Control	C2-C3	26.5	1.000	-0.16
Satisfaction	C1-C2	29.0	1.000	0.15
Satisfaction	C1-C3	10.0	0.234	-0.58
Satisfaction	C2-C3	2.0	0.012*	-1.16

Table 4: Mean (SD) SEQ (1–7), NASA-TLX composite and subscales (1–10), and comfort ratings (1–7) per modality.

Metric	M1: Text	M2: Image	M3: Voice
SEQ (ease, 1–7)	4.2 (2.0)	4.5 (2.1)	5.4 (1.6)
TLX composite	4.2 (0.6)	4.1 (0.9)	4.6 (0.6)
Mental	5.5 (2.6)	5.7 (2.9)	5.6 (2.6)
Physical	1.8 (0.9)	1.8 (0.9)	2.0 (0.8)
Temporal	2.6 (1.4)	2.8 (1.7)	5.0 (1.6)
Performance	7.3 (1.5)	6.6 (1.5)	5.8 (2.1)
Effort	3.7 (1.9)	4.2 (2.1)	5.6 (1.8)
Frustration	4.5 (1.6)	3.5 (1.5)	3.4 (2.1)
Felt natural (1–7)	4.2 (1.1)	4.9 (1.7)	4.1 (2.0)
OK in public (1–7)	4.0 (1.6)	3.3 (1.4)	2.8 (1.3)

Table 4 reports on RQ2. blah blah blah

One-sample Wilcoxon tests against the scale midpoint (4) showed that Controllability ($W = 0$, $p = .001$), Effectiveness ($W = 6$, $p = .016$), and Ownership ($W = 9$, $p = .036$) all rated significantly above neutral; Discoverability ($p = .545$) and Adoption ($p = .717$) did not differ from neutral. Controllability (agent on/off) scored highest (5.7 (1.0)), while constitution Effectiveness was lower (5.2 (1.4)), indicating that users could operate the controls but were less certain that edits changed system behaviour as expected. Discoverability (3.8 (1.6)) was the lowest-rated item, reflecting that the constitution editor remained hard to find without guidance. Overall trust in the system was 5.1 (1.4) and adoption intent was 5.0 (2.0). The most trusted feature was *Live execution graph (seeing agents work in real time)*; the most cited friction point was *Lack of undo functionality*.

Table 5: Mean (SD) customisation items (1–7 Likert), RQ3.

Item	Mean (SD)
Discoverability	3.8 (1.6)
Mental Model	4.5 (1.8)
Effectiveness	5.2 (1.4)
Controllability	5.7 (1.0)
Ownership	4.8 (1.1)
Adoption	4.2 (1.5)

4.2 Qualitative Assessment

Analysis of think-aloud and interviews was grouped into the following:

Intervention is better than observation. Participants valued the ability to *act*—approving, rejecting, or modifying a plan—more than passively watching execution. One participant noted: "Seeing the graph is cool, but what actually makes me trust it is that I can stop it before it sends the Slack message." This directly validates Amershi et al.'s [1] G2 ("support efficient correction") and G18 ("support efficient dismissal"). The quantitative data reflects this: control scores rose sharply from C1 (3.2) to C2 (5.4) (Table 2) even though C2 adds only a plan card, not a live graph—suggesting that the *ability to intervene* matters more than execution visibility.

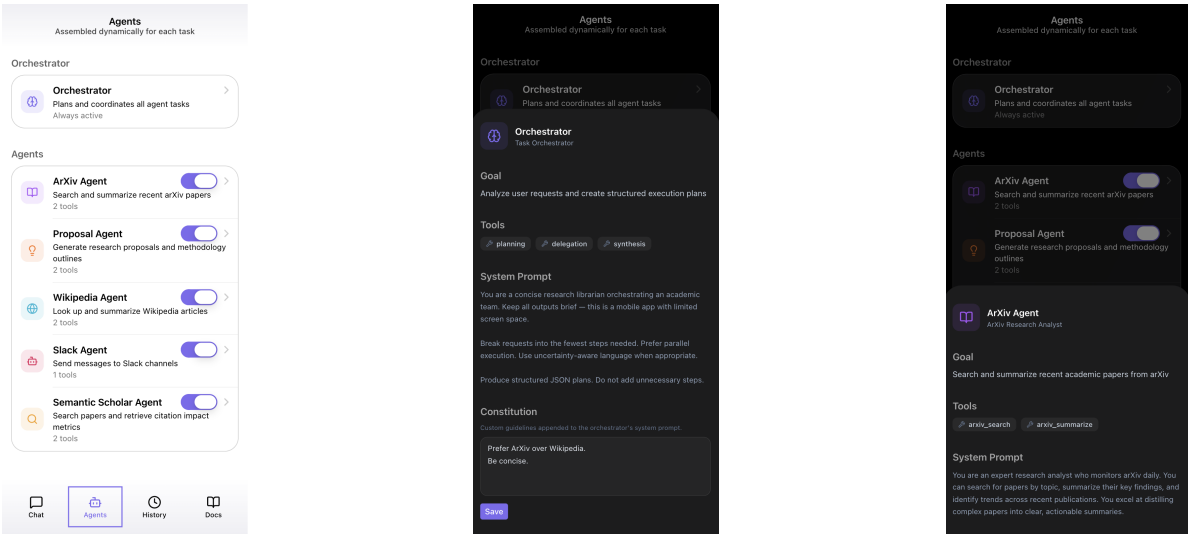
High visibility is valued

While the participants expressed enthusiasm about AI capabilities, the trust in their behavior is still low. Thus, the high level of observability in transparency mode is valued, especially in tasks that involve irreversibile actions like sending out Slack messsages (Figure 5a). This is also supported by the quantative assessment in RQ1 (Table 2; Table 3).

Theme 3: Multimodal enables mobile Participants strongly valued voice for on-the-go use but consistently avoided it in shared or public spaces. Furthermore, images enable capture of the environment that would not be possible to easily and quickly describe in text. Text remains useful as the backup modality for more complex tasks.

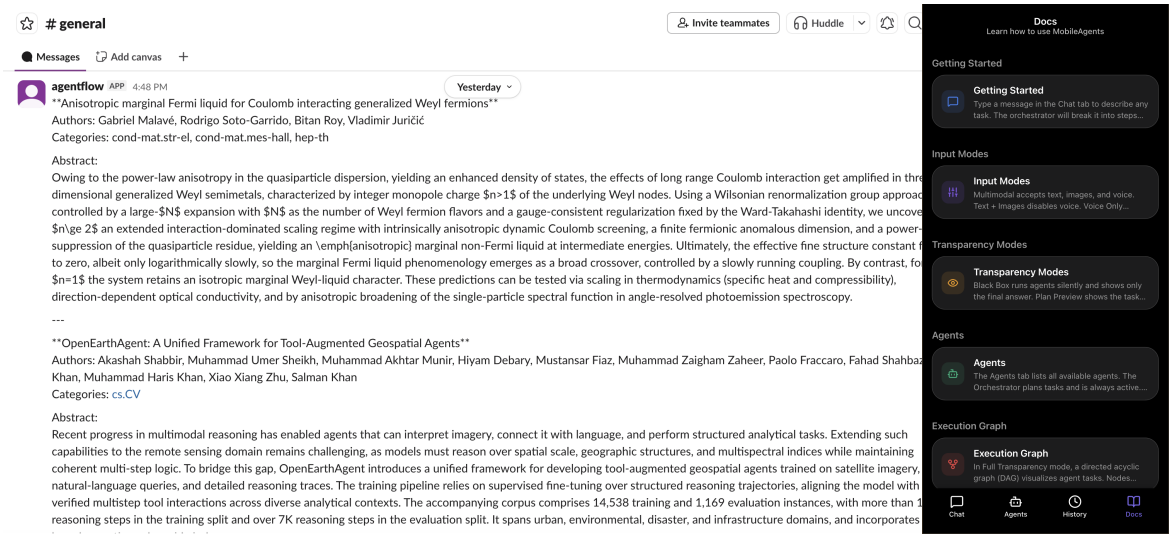
Theme 4: Constitution is confusing; agent toggles are not. Users struggled with using the constitution feature because it is difficult to discover and not immediately clear how to use (Figure 5b). However, they found the agent toggles simple, easy, and effective to build teams.

4.3 Sample Interaction Excerpts



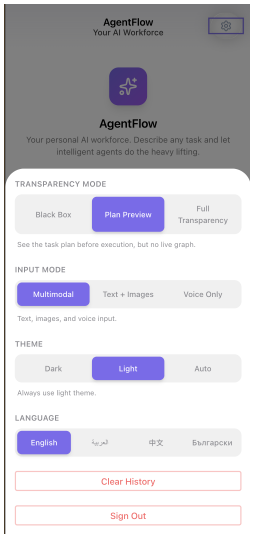
(a) Agents tab (light theme) (b) Orchestrator detail with constitution editor (c) Agent card (ArXiv agent, dark theme)

Figure 4: Agent management screens. (a) The Agents tab lists the orchestrator (always active) and all registered worker agents with per-agent enable/disable toggles. (b) Tapping the orchestrator opens its detail sheet, showing goal, tools, system prompt, and the editable constitution field. (c) A worker agent card shows its goal, tool names, and full system prompt backstory.

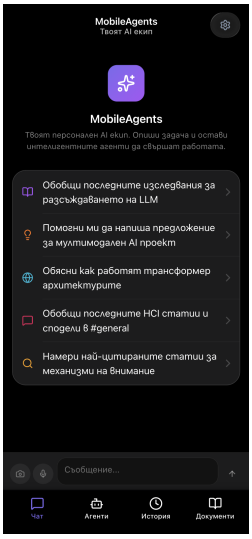


(a) Slack output (#general channel) (b) In-app documentation tab

Figure 5: Slack integration and in-app documentation. (a) Result of a Slack send step after checkpoint approval: two arXiv paper summaries posted to #general by the agentflow bot. (b) The Docs tab provides an in-app guide covering input modes, transparency modes, the agents tab, and execution graph; it is available without leaving the app.



(a) Settings panel



(b) Bulgarian localisation

Figure 6: Settings and localisation screens. (a) Settings exposes all three transparency modes, input modes (Multimodal / Text+Images / Voice Only), theme (Dark/Light/Auto), and language (EN/AR/ZH/BG). (b) Bulgarian localisation: the full UI—including navigation labels and suggested queries—renders in Bulgarian with full RTL support for Arabic (not shown).

5 Discussion Future Work

5.1 Ethical, Social, and Value-Related Reflection

Automation vs. autonomy on mobile.

Our findings surface the difficulty of designing an agentic application that prioritizes both efficiency and control. Users want agents to be autonomous but do not want to give up control. Black Box scores lowest on trust and control. Yet, certain users preferred it for low-stakes queries. This issue is especially critical in mobile settings, as each additional tap undermines the motivation for ambient, on-the-go assistance. The transparency mode received the most trust. However, we recognize that the live graph may create an illusion of control and oversight. Future designs can explore an auto mode that can classify individual prompts into the appropriate control setting.

Privacy Security

We recognize that a single centralized agentic manager brings risks for data storage and prompt injection from malicious agents. Thus, we release the tool as an open-source solution so that individual users and organizations can self-host on-premises and own all of their data. The current system is vulnerable to adversarial attacks from the integrated agents. In the current version, we recommend bringing only trusted agents. Future work should research the risks and integrate guardrails and safety mechanisms to prevent unintended behavior.

Inclusivity and access.

Mobile Agents is built around inclusivity and broad access through deep customization. However, our design and testing have been limited in scope and size and is not currently representative of the wide range of potential users. Users with lower technical literacy may struggle to utilize our architecture. Certain features like the constitution and bring-your-own agent archiecture is currently only accessible to power users. While we have a tab with documentation, it should be expanded to cover the topics in simpler, more accessible language. We should support easier integration with agents through MCP servers and expand current accessibility coverage.

Conclusion. MobileAgents demonstrates that personal AI agents can be made controllable from a mobile device through multimodal input (text, voice, camera) and an agent-agnostic orchestration layer. The bring-your-own-agents architecture means users are not locked into a fixed agent set—they plug in whatever tools matter to them and manage them from their phone. Configurable transparency lets users match oversight to context: Plan Preview for quick mobile use, Full Transparency for high-stakes actions. The ability to *intervene*—not merely observe—drove trust more than visual complexity. Consent-gated multimodality respected privacy in mobile contexts, though speech recognition equity and automation bias remain open challenges. These findings contribute to the design space of mobile, multimodal, user-controllable multi-agent AI systems.

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A Project Structure

```
agentflow/
+-- package.json          # Monorepo scripts
+-- client/
|   \-- src/
|       +-- App.tsx       # Root (auth, theme, RTL)
|       +-- index.css     # Tailwind + animations
|       +-- state/
|       |   \-- store.ts  # Zustand (state + API)
|       +-- types/        # TS interfaces
|       +-- i18n/         # 4 languages (en/zh/ar/bg)
|       \-- components/
|           +-- Chat/     # Chat + markdown + TTS
|           +-- Agents/   # List + detail + toggle
|           +-- Graph/    # ReactFlow DAG + nodes
|           +-- TaskPlan/ # Plan card + approval
|           +-- Camera/   # Image capture + consent
|           +-- Voice/    # Recording + consent
|           +-- Settings/ # Transparency/theme/i18n
|           +-- Execution/# History + step stream
|           \-- Privacy/  # Consent dialog
+-- server/
|   +-- main.py           # FastAPI + CORS + routers
|   +-- config.py         # Pydantic settings
|   +-- auth.py           # Supabase JWT auth
|   +-- routers/
|   |   +-- chat.py       # POST /api/chat
|   |   +-- execute.py    # POST /api/execute (SSE)
|   |   +-- approve.py    # Checkpoint approval
|   |   +-- agents.py     # Agent CRUD
|   |   \-- conversations.py # Conversation CRUD
|   +-- crew/
|   |   +-- orchestrator.py # Planning + graph
|   |   +-- agents.py      # Backstories
|   |   \-- tools.py       # Tool implementations
|   +-- models/
|   |   +-- messages.py   # ChatRequest/Response
|   |   +-- tasks.py      # TaskPlan/TaskStep
|   |   \-- graph.py      # Graph state models
|   \-- services/
|       +-- execution_tracker.py # Dependency exec
|       +-- image_analyzer.py    # GPT-4.1 vision
|       +-- audio_transcriber.py # Whisper
|       +-- agent_store.py       # JSON agent CRUD
|       +-- db.py                # Supabase DB layer
|       \-- supabase_client.py   # Supabase client
```

B System Prompts

B.1 Orchestrator Backstory

```
You are a concise research librarian orchestrating
an academic team. Keep all outputs brief — this is
a mobile app with limited screen space.

Break requests into the fewest steps needed.
Prefer parallel execution.
Use uncertainty-aware language when appropriate.

Produce structured JSON plans. Do not add
unnecessary steps.

Available agents:
- arxiv: ArXiv Research Analyst —
    capabilities: arxiv_search, arxiv_summarize
- proposal: Research Proposal Strategist —
    capabilities: generate_proposal,
    outline_methodology
- wikipedia: Background Research Specialist —
    capabilities: wiki_search, wiki_summarize
- slack: Slack Messenger —
    capabilities: slack_send_message
    [requires_approval]
- semantic_scholar: Citation Analyst —
    capabilities: semantic_scholar_search,
    semantic_scholar_cite

Constitution (user-defined guidelines):
<user-editable free text appended here>
```

B.2 Planning Task Prompt

```
Analyze the following user request and create a
structured execution plan.

User request: {user_message}

[Image analysis: {image_analysis}]
[Audio transcript: {audio_transcript}]
[Conversation history: {history}]

Input modality: {input_modality}
{modality_note}

Available agents:
{agent_descriptions}

Rules:
- Match the user's request to the most appropriate
  agent(s). Agents are NOT limited to research —
  some send messages, perform actions, etc.
- For agents marked [requires_approval], set
  requires_approval=true on that step. For all
  other agents, set requires_approval=false.
- Only return an empty steps list if the request
  truly cannot be handled by ANY available agent
  (e.g. "hello", pure casual chat).
- If steps are independent, leave depends_on empty
  so they can run in parallel.
- If a step needs output from another step, add
  that step's ID to depends_on.
- Always prefer parallel execution when possible.
- Use step IDs like "step_1", "step_2", etc.
- agent_id must be one of: {valid_agent_ids}
- Use exact param names from tool signatures.
  For slack_send_message use params:
  {"channel": "<channel>", "text": "<message>"}
- If image analysis is provided, use that content
  to inform search queries and proposal topics.
- If audio was transcribed, treat the transcript
  as the primary user intent.
- If conversation history is provided, use it to
  resolve ambiguous references.
```

B.3 Synthesis Prompt

System prompt:

```
You are a research orchestrator synthesizing agent
results for a MOBILE interface. Responses are read
on small screens.

Guidelines:
- Be concise: aim for 150-300 words max
- Lead with the key finding or answer in 1-2
  sentences
- Use short bullet points, not long paragraphs
- Use markdown: **bold** for emphasis, short
  headings, compact lists
- Keep relevant URLs but don't list more than 3-5
- Do NOT mention the agents or internal execution
  process
- Use uncertainty-aware language: 'based on
  available results' rather than absolute claims
- End with a one-line disclaimer: '*Results may
  not be exhaustive — verify for critical
  decisions.*'
```

User message template:

```
**User request:** {user_message}

**Plan summary:** {plan_summary}

**Agent results:**

### Agent step: {step_description}
{step_result}

### Agent step: {step_description}
{step_result}
...
```

B.4 Image Analysis Prompt

```
Analyze this image in the context of a research
assistant. Describe what you see and what research
tasks might be relevant (e.g., paper search,
literature review, proposal generation).
Be concise.
```

B.5 Modality Notes

C Agent Backstories

```
[ArXiv Agent]
You are an expert research analyst who monitors arXiv
daily. You can search for papers by topic, summarize
```

Table 6: Modality notes injected into the planning prompt.

Modality	Injected note
text	"The user typed this request."
voice	"The user spoke this request via voice input."
image	"The user provided an image along with their request. The image has been analyzed and the analysis is included above."

```
their key findings, and identify trends across recent
publications. You excel at distilling complex papers
into clear, actionable summaries.

[Proposal Agent]
You are a seasoned research strategist who has helped
write dozens of successful grant proposals and research
plans. You understand how to identify research gaps,
formulate compelling research questions, and outline
rigorous methodologies. You synthesize information from
literature into coherent proposals.

[Wikipedia Agent]
You are a thorough background researcher who uses
Wikipedia to gather foundational knowledge on topics.
You find relevant articles, extract key concepts and
definitions, and provide well-structured overviews that
help contextualize research work.

[Slack Agent]
You are a communication specialist who sends messages
to Slack channels on behalf of the team. You craft
clear, well-formatted messages and deliver them to the
correct channels. You always confirm which channel to
post to and summarize what was sent.

[Semantic Scholar Agent]
You are a citation analyst who uses Semantic Scholar to
find papers and assess their impact. You search across
all academic literature (not just arXiv), retrieve
citation counts and influential citation metrics, and
help identify the most impactful work in any research
area.
```

D Tool Implementations

D.1 SSE Event Protocol

Execution updates are streamed as Server-Sent Events.

E Data Schemas

E.1 Request and Response Models

ChatRequest (client → server):

```
{
  "message": "string",
  "image_base64": "string | null",
  "audio_base64": "string | null",
  "input_modality": "text | voice | image",
  "conversation_history": [
    {"role": "user|assistant", "content": "..."}
  ],
  "conversation_id": "string | null"
}
```

ChatResponse (server → client):

```
{
  "message": "string",
  "plan": TaskPlan | null,
  "graph": ExecutionGraphState | null,
  "image_base64": "string | null"
}
```

E.2 Task Plan Models

TaskPlan:

```
{
  "id": "string",
  "summary": "string",
  "steps": [TaskStep],
  "status": "proposed | executing | completed",
  "created_at": "ISO 8601"
}
```

TaskStep:

```
{
  "id": "step_1",
  "agent_id": "arxiv | proposal | wikipedia
              | slack | semantic_scholar",
  "action": "tool function name",
  "description": "human-readable description",
  "params": { "key": "value" },
  "status": "pending | running | completed
            | failed | awaiting_approval",
  "result": "string | null",
  "requires_approval": false,
  "depends_on": ["step_id"]
}
```

E.3 Execution Graph Models

ExecutionGraphState:

```
{
  "task_id": "string",
  "nodes": [GraphNode],
  "edges": [GraphEdge],
  "current_node_id": "string | null",
  "status": "planning | executing
            | completed | failed"
}
```

GraphNode:

```
{
  "id": "string",
  "type": "input | orchestrator | agent
          | checkpoint | output",
  "data": {
    "label": "string",
    "type": "...",
    "status": "pending | running | completed
              | failed | awaiting_approval
              | approved | skipped",
    "agent_id": "string | null",
    "agent_color": "#hex | null",
    "agent_icon": "LucideIconName | null",
    "result": "string | null",
    "duration": "milliseconds | null",
```

Table 7: Complete tool registry. Each function is mapped to one or more agents and called directly during execution.

Function	Agent	Parameters	Behaviour
`arxiv_search`	ArXiv	`query`: str, `max_results`: int (default 5)	Queries the arXiv API sorted by submission date (descending). Returns titles, authors, URLs, and truncated abstracts (300 chars). Timeout: 15 s.
`arxiv_summarize`	ArXiv	`paper_url`: str	Fetches a single paper by arXiv ID. Returns title, authors, categories, and full abstract.
`generate_proposal`	Proposal	`topic`: str, `context`: str	Generates a Markdown research proposal with sections: Introduction, Research Questions, Methodology, Expected Contributions, Timeline. Context from prior steps is incorporated.
`outline_methodology`	Proposal	`approach`: str, `domain`: str	Generates a methodology section with Data Collection, Approach Details, Evaluation Metrics, and Reproducibility.
`wiki_search`	Wikipedia	`query`: str	Searches the Wikipedia API; returns up to 5 results with snippets. User-Agent: `MobileAgents/1.0`. Timeout: 10 s.
`wiki_summarize`	Wikipedia	`title`: str	Fetches the Wikipedia REST API summary for a page. Returns title, URL, and extract.
`slack_send_message`	Slack	`channel`: str, `text`: str	POSTs to the Slack API with a Bearer token (`SLACK_BOT_TOKEN`). Returns confirmation or error. Requires checkpoint approval. Timeout: 10 s.
`semantic_scholar_search`	Sem. Scholar	`query`: str, `max_results`: int (default 5)	Queries the Semantic Scholar API. Returns title, year, citation count, influential citation count, authors, URL, and abstract. Timeout: 15 s.
`semantic_scholar_cite`	Sem. Scholar	`paper_id`: str	Fetches citation details for a paper by Semantic Scholar ID, DOI, or `ARXIV:<id>`. Returns citation count, influential citation count, reference count, authors. Handles 404 gracefully. Timeout: 15 s.

Table 8: SSE event types streamed during execution.

Event type	Payload and semantics
`graph_init`	Full `ExecutionGraphState`. Sent once at the start.
`node_status`	`nodeId`, `status` (`running` | `completed` | `failed` | `awaiting_approval` | `approved`), optional `result` and `duration` (ms).
`edge_status`	`edgeId`, `status` (`active` | `completed`).
`checkpoint_reached`	`nodeId`, `stepId`. Emitted when an approval-gated step is ready.
`execution_complete`	Updated `ExecutionGraphState` and synthesised `summary`.
`execution_failed`	`error` message string.

```
    "input_modality": "text|voice|image | null",
    "timestamp": "ISO 8601 | null"
  },
  "position": { "x": 0, "y": 0 }
}
```

GraphEdge:

```
{
  "id": "e-source-target",
  "source": "node_id",
  "target": "node_id",
  "data": {
    "status": "pending | active | completed",
    "data_preview": "string | null"
  }
}
```

F Agent Registry Schema

```
{
  "id": "unique_string",
  "name": "Display Name",
  "icon": "LucideIconName",
  "description": "Short description",
  "role": "CrewAI role string",
  "goal": "CrewAI goal string",
  "backstory": "Full system prompt text",
  "capabilities": ["tool_1", "tool_2"],
  "enabled": true,
  "requiresApproval": false,
  "color": "#hex",
  "isOrchestrator": false,
  "constitution": ""
}
```

G CSS Animations and Theming

G.1 Theme Variables

Table 9: Theme colour tokens (selected).

Token	Light	Dark
`-surface-0`	`#ffffff`	`#000000`
`-surface-1`	`#f2f2f7`	`#1c1c1d`
`-surface-2`	`#e5e5ea`	`#121212`
`-text-primary`	`#000000`	`#ffffff`
`-text-secondary`	`#3c3c43`	`#ffffffcc`

H Authentication and Data Persistence

H.1 Authentication

The server uses Supabase for authentication.

(1) Extracts the Bearer token from the `Authorization` header.
(2) Validates it via `supabase.auth.get_user(token)`, which works with both HS256 and ES256 JWT keys.

(3) Returns the following object or raises HTTP 401:

```
{"sub": "user_id", "email": "user_email"}
```

The client stores the Supabase session token and includes it in all API requests.

H.2 Server-Side Persistence

Conversations, messages, execution records, and approval decisions are persisted to Supabase (PostgreSQL). The database layer (`services/db.py`) provides:

- **Conversations:** Create, list (by user), get, delete. Each conversation stores a title and transparency level.
- **Messages:** Insert and list (by conversation). Each message stores role, content, optional image URL, voice transcript, agent ID, input modality, task plan, and execution graph.
- **Executions:** Create (with plan and graph snapshots) and complete (with status, summary, and step results).
- **Approvals:** Create, resolve (approve/reject with optional comment), and query by step ID.

I Survey Instrument

All Likert items use a 7-point scale (1 = Strongly Disagree, 7 = Strongly Agree) unless otherwise noted.

Part 0: Background (before study begins)

(1) Age range. *(18–21 / 22–25 / 26–30 / 31–40 / 40+)*
(2) Gender. *(Male / Female / Non-binary / Prefer not to say)*
(3) How often do you use AI assistants (e.g., ChatGPT, Copilot)? *(Daily / Several times a week / Weekly / Monthly / Rarely or Never)*
(4) Have you used a multi-agent AI system before? *(Yes / No / I'm not sure)*
(5) Which AI tools do you use regularly? *(Select all: ChatGPT, Copilot, Gemini, Claude, Cursor, Perplexity, Other)*

Part 1: After each transparency condition (C1, C2, C3) — RQ1

Completed three times, once per configuration. Items 1–3 are primary metrics; items 4–7 are supplementary.

(1) I trusted the system's output enough to act on it. *(Trust)*
(2) I felt in control of what the system did on my behalf. *(Perceived Control)*
(3) I am satisfied with this interaction. *(Satisfaction)*
(4) The system made it clear what it was going to do before doing it. *(Expectation Setting)*
(5) It was easy to correct or stop the system when needed. *(Repairability)*
(6) I understood why the system made the decisions it did. *(Explainability)*
(7) I felt comfortable with the information I shared during this task. *(Privacy Comfort)*
– Any comments about this mode? *(open text, optional)*

After all three conditions:

8. Which transparency mode would you prefer for everyday use? *(C1 Blackbox / C2 Plan Preview / C3 Full Transparency)*
9. Which mode would you prefer for high-stakes tasks (e.g., sending a Slack message on your behalf)? *(C1 / C2 / C3)*
10. Why did you prefer that mode? *(open text, optional)*

Part 2: After each modality condition (M1, M2, M3) — RQ2

Completed three times, once per modality (text, image, voice). The NASA-TLX uses a 1–10 scale (Very Low–Very High); Performance is reversed (1 = Perfect, 10 = Failure).

Single Ease Question (SEQ):

(1) Overall, how would you rate the difficulty of this task? *(1 = Very Difficult, 7 = Very Easy)*

NASA Task Load Index [?]:

2. Mental Demand — How much mental and perceptual activity was required? *(1–10)*
3. Physical Demand — How much physical activity was required? *(1–10)*
4. Temporal Demand — How much time pressure did you feel? *(1–10)*
5. Performance — How successful were you in accomplishing the task? *(1 = Perfect, 10 = Failure)*
6. Effort — How hard did you have to work to accomplish the task? *(1–10)*
7. Frustration — How irritated, stressed, or annoyed did you feel? *(1–10)*

Modality comfort (7-point Likert):

8. Using this modality felt natural for this type of task. *(Modality Comfort)*
9. I would use this modality in a public or shared space. *(Context Appropriateness)*
– In what situations would you choose or avoid this modality? *(open text, optional)*

After all three modality conditions:

10. Which input modality would you use most often on a mobile device? *(Text / Voice / Image / It depends)*
11. Which modalities felt well-suited to on-the-go use? *(Select all: Text / Voice / Image)*
– Was there a modality that surprised you – positively or negatively? *(open text, optional)*

Part 3: After the customisation task (Phase 3) — RQ3

(1) I was able to find the constitution editor without help. *(Discoverability)*
(2) I understood what the constitution field does. *(Mental Model)*
(3) Editing the constitution changed the system's behaviour as I expected. *(Effectiveness)*
(4) The agent on/off toggles were easy to understand and use. *(Controllability)*
(5) After customising the system, I felt more in control. *(Perceived Ownership)*
(6) I would use the constitution and agent controls regularly. *(Adoption Intent)*
7. Which type of change did you make to the constitution? *(Constraint / Style / Routing / Did not edit / Other)*

- Describe what you changed and whether the system behaved as expected. *(open text)*

Part 4: Overall post-study feedback

(1) Overall, I would trust this system to act on my behalf on a mobile device. *(Overall Trust)*
(2) I felt this system respected my privacy. *(Privacy)*
(3) I would use this application in my day-to-day work. *(Adoption Intent)*

4. Which single feature contributed most to your trust? *(Plan Preview / Checkpoint approval / Live DAG / Agent on/off / Constitution / Multimodal input)*
5. Which single feature caused the most friction or concern? *(Plan approval steps / Live graph overwhelming / Voice risky in shared spaces / Camera awkward / Agent controls hard to find / Constitution unclear)*

- What is the single most important improvement you would make? *(open text)*
- Any other comments or suggestions? *(open text)*

J Semi-Structured Interview Questions

After completing all three configurations, participants were asked:

(1) You used three different versions of the system. Can you describe how each felt different?
(2) Which version would you prefer for everyday use? Why?
(3) Was there a moment where you particularly trusted—or distrusted—the system? What caused that?
(4) The system showed you a plan before executing. Did that change how you interacted with it? How?
(5) For the version with the live graph, was seeing the execution progress helpful or distracting?
(6) The system required approval before sending a Slack message. How did that approval step feel? Too cautious? About right?
(7) How did you decide whether to use text, voice, or camera input? Were there situations where you avoided a modality? Why?
(8) Did any of the system's responses feel overconfident or uncertain? How did that affect your trust?
(9) If this system stored your conversation history across sessions, how would you feel about that? What controls would you want?
(10) Can you think of a situation where a system like this could cause harm—to you or someone else?
(11) What is the single most important change you would make to improve this system?